INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 10

1. Experiment Title:

Joining Two Tables with INNER JOIN

2. Aim:

Join department and employee tables using their common dept_id column.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

A JOIN combines related records from two or more tables. INNER JOIN returns only rows that satisfy the join condition. Table aliases can make queries shorter and easier to read.

5. Algorithm / Procedure:

1. Create department and employee tables and insert the supplied records.
2. Join employee to department on dept_id and sort by employee name.
3. Filter the joined rows for the Sales department.
4. Display employee name, department name, and salary in descending salary order.

6. Program / Queries / Commands:

```
CREATE TABLE department (
    dept_id INT,
    dept_name VARCHAR(20)
);

INSERT INTO department (dept_id, dept_name)
VALUES (10, 'Accounts'), (20, 'Sales'), (40, 'Research');

CREATE TABLE employee (
    emp_name VARCHAR(30),
    dept_id INT,
    salary INT
);

INSERT INTO employee (emp_name, dept_id, salary)
VALUES
    ('Anita', 10, 90000),
    ('Lata', 20, 55000),
    ('Deepak', 10, 60000),
    ('Pooja', 20, 42000);

-- Task 1
SELECT e.emp_name, d.dept_name
FROM employee AS e
INNER JOIN department AS d
    ON e.dept_id = d.dept_id
ORDER BY e.emp_name;

-- Task 2
SELECT e.emp_name
FROM employee AS e
INNER JOIN department AS d
```

```
    ON e.dept_id = d.dept_id
WHERE d.dept_name = 'Sales'
ORDER BY e.emp_name;

-- Task 3
SELECT e.emp_name, d.dept_name, e.salary
FROM employee AS e
INNER JOIN department AS d
    ON e.dept_id = d.dept_id
ORDER BY e.salary DESC;
```

7. Input:

Departments: (10, Accounts), (20, Sales), (40, Research). Employees: Anita/10/90000, Lata/20/55000, Deepak/10/60000, Pooja/20/42000.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: emp_name	**dept_name**	
Anita	Accounts	
Deepak	Accounts	
Lata	Sales	
Pooja	Sales	
Task 2: emp_name		
Lata		
Pooja		
Task 3: emp_name	dept_name	salary
Anita	Accounts	90000
Deepak	Accounts	60000
Lata	Sales	55000
Pooja	Sales	42000

9. Result:

The SQL statements for Experiment 10 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is a join in SQL?
2. What does INNER JOIN return?
3. Why is dept_id used in the ON condition?
4. What is a table alias?